Table 1: What each conditioner can express at each budget (`hyperkkl.conditioning.match_width` on the phase-1 decoder shape, dec. width 128, depth 3, context dim 16). The dense-subspace ablation ('hyper') generates every weight update inside a FIXED k-dimensional linear subspace (k = trunk output width); HyperKKL ('lora') has rank $\leq r$ per layer with factors bilinear in a w-dim trunk feature. Observed: widths in the runs' results.json (– = no run yet).

system (n_z)	budget	variant	width	params	max. rank per layer	trunk	update	obser
vdp (5)	35k	HyperKKL $r = 4$	11	35,953	[4, 4, 4, 2]	11-dim feature	$U_l(c)V_l(c)$, bilinear	
vdp (5)	35k	dense (abl.)	1	35,138	–	$16 \to 64 \to 1$	ΔW in a fixed 1-dim subspace (34,050×1 heads)	
vdp (5)	140k	HyperKKL $r = 1$	149	140,449	[1, 1, 1, 1]	149-dim feature	$U_l(c)V_l(c)$, bilinear	
vdp (5)	140k	HyperKKL $r = 4$	44	140,596	[4, 4, 4, 2]	44-dim feature	$U_l(c)V_l(c)$, bilinear	
vdp (5)	140k	HyperKKL $r = 16$	11	142,921	[5, 16, 16, 2]	11-dim feature	$U_l(c)V_l(c)$, bilinear	
vdp (5)	140k	dense (abl.)	4	137,480	–	$16 \to 64 \to 4$	ΔW in a fixed 4-dim subspace (34,050×4 heads)	
vdp (5)	560k	HyperKKL $r = 4$	170	560,176	[4, 4, 4, 2]	170-dim feature	$U_l(c)V_l(c)$, bilinear	
vdp (5)	560k	dense (abl.)	16	546,848	–	$16 \to 64 \to 16$	ΔW in a fixed 16-dim subspace (34,050×16 heads)	
battery (7)	35k	HyperKKL $r = 4$	11	36,181	[4, 4, 4, 4]	11-dim feature	$U_l(c)V_l(c)$, bilinear	
battery (7)	35k	dense (abl.)	1	35,781	–	$16 \to 64 \to 1$	ΔW in a fixed 1-dim subspace (34,693×1 heads)	
battery (7)	140k	HyperKKL $r = 1$	148	140,103	[1, 1, 1, 1]	148-dim feature	$U_l(c)V_l(c)$, bilinear	
battery (7)	140k	HyperKKL $r = 4$	44	141,484	[4, 4, 4, 4]	44-dim feature	$U_l(c)V_l(c)$, bilinear	
battery (7)	140k	HyperKKL $r = 16$	11	143,833	[7, 16, 16, 5]	11-dim feature	$U_l(c)V_l(c)$, bilinear	
battery (7)	140k	dense (abl.)	4	140,052	–	$16 \to 64 \to 4$	ΔW in a fixed 4-dim subspace (34,693×4 heads)	
battery (7)	560k	HyperKKL $r = 4$	169	560,109	[4, 4, 4, 4]	169-dim feature	$U_l(c)V_l(c)$, bilinear	
battery (7)	560k	dense (abl.)	16	557,136	–	$16 \to 64 \to 16$	ΔW in a fixed 16-dim subspace (34,693×16 heads)	